

RUT Perturbation Sprint

From Action to E_G Falsifier

Compiled technical packet

Reverse Universe Theory (RUT)

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This packet consolidates the perturbation sprint into a single technical trail: conventions, field equations, FRW reduction, bridge perturbations, linearized Einstein equations, quasi-static matrix reduction, gravitational slip, the E_G falsifier, parameter discipline, and unresolved derivation targets. It is intended as a working research packet, not a claim that all coefficients have been fully derived.

Claim-level rule: use [D] for derived results, [D/M] for structurally derived but model-dependent results, [A/M] for working ansatz/parameterization, [O] for open derivation targets, and [I] for interpretive material.

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Total packet length: 34 pages, including cover and table of contents.

RUT Perturbation Sprint - Step 1

Fixing conventions for metric signature, stress-energy, Newtonian gauge, modified gravity functions, and slip

Purpose. This step establishes the conventions and definitions that will be used for all subsequent perturbation-level derivations in RUT. These are definitions, not derivations, and are labelled accordingly as [I] input conventions.

Metric Signature [I]

Adopt the mostly-plus convention throughout:

$$(-, +, +, +).$$

All contractions, stress-energy signs, and field-equation variations in this sprint should be checked against this metric signature.

Stress-Energy Tensor [I]

Define the energy-momentum tensor for matter and field sectors by:

$$T_{\{\mu\nu\}} = -2 / \sqrt{-g} \delta S_{\{\text{matter/fields}\}} / \delta g^{\{\mu\nu\}}.$$

This convention keeps the sector stress-energy tensors consistent with the Einstein equations used in the RUT action-level framework.

Coordinate Gauge [I]

Work in Newtonian, also called longitudinal, gauge for scalar perturbations:

$$ds^2 = -(1+2\Psi)dt^2 + a^2(t)(1-2\Phi)d x^2.$$

Here Ψ and Φ are the Bardeen potentials. Ψ controls the nonrelativistic force potential, while Φ contributes with Ψ to the Weyl/lensing potential.

Modified-Gravity Definitions [I]

Define the scale- and time-dependent functions $\mu(k,a)$ and $\Sigma(k,a)$ operationally through the potentials that control growth and lensing:

$$\begin{aligned} -k^2 \Psi &= 4\pi G a^2 \mu(k,a) \bar{\rho}_m \delta_m, \\ -k^2 (\Phi + \Psi) &= 8\pi G a^2 \Sigma(k,a) \bar{\rho}_m \delta_m. \end{aligned}$$

Here $\bar{\rho}_m$ is the background matter density and δ_m is the matter overdensity. In general relativity, under the standard no-anisotropic-stress limit and with these operational definitions, $\mu = \Sigma = 1$.

Gravitational Slip [I]

Define the slip or potential ratio as:

$$\eta(k,a) = \Phi / \Psi.$$

Using the modified-gravity definitions above, it follows that:

$$\eta(k,a) = 2 \Sigma(k,a) / \mu(k,a) - 1.$$

In GR, where $\mu = \Sigma = 1$, this gives $\eta = 1$. This convention avoids the earlier sign-confusion problem by defining μ and Σ first and deriving slip from them second.

Convention rule for the sprint

Use $\mu(k,a)$ for growth, $\Sigma(k,a)$ for lensing/Weyl response, and $\eta(k,a)=\Phi/\Psi$ only as a derived secondary slip variable. This keeps the E_G pipeline convention-safe.

Next step. With these conventions fixed, Step 2 can safely derive the RUT perturbation equations from the action without changing signs or redefining slip midway through the calculation.

RUT Perturbation Sprint - Step 2

Schematic Field Equations from the Action

Below are the field equations obtained by varying the RUT action with respect to each field. Because these equations arise directly from functional derivatives of the action, they are labelled **[D]**. Exact numerical factors and signs should still be checked in the detailed derivation, but the structure shown here follows from the calculus of variations.

2.1 Einstein Equations

Varying the action with respect to the metric $g_{\mu\nu}$ yields the modified Einstein equations:

$$M_{Pl}^2 G_{\{\mu\nu\}} = T_{\{\mu\nu\}}^{\{(m)\}} + T_{\{\mu\nu\}}^{\{(\phi)\}} + T_{\{\mu\nu\}}^{\{(\chi)\}} + T_{\{\mu\nu\}}^{\{(A)\}} + T_{\{\mu\nu\}}^{\{(Q)\}}.$$

Each stress-energy tensor comes from a sector of the action. The scalar, bridge, gauge, and Q-sector stress-energy tensors will be specified after the sign check in Step 3. **[D]**

2.2 Scalar Phase Field ϕ

Variation of the ϕ -dependent terms

$$-1/2 K(B)(\nabla\phi)^2 - V(\phi, B) - 1/2 \xi_Q B(\chi) (\nabla^\mu\theta \nabla_\mu\phi)^2$$

leads, after integration by parts, to the equation of motion:

$$\nabla_\mu [K(B) \nabla^\mu\phi] - V_{,\phi} + \nabla_\mu [\xi_Q B(\chi) (\nabla^\alpha\theta \nabla_\alpha\phi) \nabla^\mu\theta] = 0.$$

The first term comes from varying the kinetic term, the second from the potential, and the third from the Q-sector coupling. **[D]**

2.3 Bridge Field χ

Varying the bridge kinetic term, the bridge potential, and all terms that depend on $B(\chi)$ gives:

$$B_{,\chi} - U_{,\chi} + B_{,\chi} [-1/2 K_{,B} (\nabla\phi)^2 - V_{,B} - 1/4 Z_{,B} F_{\{\mu\nu\}} F^{\{\mu\nu\}} - 1/2 \xi_Q (\nabla^\alpha\theta \nabla_\alpha\phi)^2] = 0.$$

where

$$B_{,\chi} = [1/(2 \Delta\chi)] \operatorname{sech}^2((\chi - \chi_*)/\Delta\chi).$$

The terms inside the brackets arise from differentiating $K(B)$, $V(\phi, B)$, $Z(B)$, and the Q-sector with respect to B . **[D]**

2.4 Gauge Field A_μ

Varying the gauge kinetic term and including any matter current J_m^μ from the matter Lagrangian gives:

$$\nabla_\mu [Z(B) F^{\{\mu\nu\}}] = J_m^\nu.$$

In the absence of external currents, $J_m^\mu = 0$. The factor $Z(B)$ modulates the electromagnetic sector through the bridge field. **[D]**

2.5 Light-Phase Field θ

The Q-sector depends on θ only through

$$X = \nabla^\mu\theta \nabla_\mu\phi.$$

Varying the Q-term with respect to θ yields a conservation-type equation:

$$\nabla_\mu [\xi_Q B(\chi) (\nabla^\alpha \theta \nabla_\alpha \phi) \nabla^\mu \phi] = 0.$$

This shows that θ is not free; its dynamics are tied to the gradient of ϕ through the bridge. **[D]**

Transition to Step 3

These equations capture the essential structure of the field dynamics needed for later steps. In Step 3, the Q-sector stress-energy tensor should be computed explicitly and the signs should be checked before proceeding to the FRW background reduction.

RUT Perturbation Sprint - Step 3

Q-sector stress-energy tensor, sign choice, and anisotropic-stress role

Below is the key result for Step 3: the Q-sector stress-energy tensor derived from the RUT action. This section is labelled **[D]** because it follows from functional variation of the action. Exact sign conventions should remain tied to the mostly-plus metric convention fixed in Step 1.

Q-sector Stress-Energy Tensor [D]

Recall the Q-sector Lagrangian:

$$L_Q = -1/2 \xi_Q B(\chi) (\nabla^\mu \theta \nabla_\mu \phi)^2$$

Define

$$Q = g^{\alpha\beta} \partial_\alpha \theta \partial_\beta \phi, \text{ so } L_Q = -1/2 \xi_Q B Q^2.$$

Varying the action with respect to the metric $g^{\mu\nu}$, while keeping ϕ , χ , and θ fixed, gives the energy-momentum tensor

$$T_{\mu\nu}^{(Q)} = -2 \partial L_Q / \partial g^{\mu\nu} + g_{\mu\nu} L_Q.$$

The metric derivative of the Q-sector is

$$\partial L_Q / \partial g^{\mu\nu} = 1/2 \xi_Q B Q (\partial_\mu \theta \partial_\nu \phi + \partial_\mu \phi \partial_\nu \theta).$$

Therefore, the Q-sector stress-energy tensor is

$$T_{\mu\nu}^{(Q)} = \xi_Q B Q (\partial_\mu \theta \partial_\nu \phi + \partial_\mu \phi \partial_\nu \theta) - 1/2 \xi_Q B Q^2 g_{\mu\nu}.$$

Sign choice

- The first term is written with the positive sign so that, on a homogeneous background where $\partial_i \theta = 0$ and $\partial_i \phi = 0$, the energy density $T_{00}^{(Q)}$ is proportional to $(\dot{\theta} \dot{\phi})^2$ and is positive when $\xi_Q B > 0$.
- Alternate sign choices can lead to negative energy density for the Q-sector. Such choices would be physically dangerous unless compensated by a deeper stability argument.

Anisotropic stress

- The off-diagonal spatial components, with $i \neq j$, contain cross-gradient terms of the form $\partial_i \theta \partial_j \phi + \partial_i \phi \partial_j \theta$.
- In a strictly homogeneous background these vanish, so the Q-sector does not produce anisotropic stress at zeroth order.
- At linear order in perturbations, the cross-gradient term can generate an anisotropic-stress source for the metric slip $\Phi - \Psi$. This contribution must be included when deriving the RUT perturbation equations and the lensing/growth functions $\mu(k,a)$, $\Sigma(k,a)$, and $E_G(k,a)$.

Step 3 conclusion

This concludes the Q-sector stress-energy step. The next step is to use this result to compute the FRW background quantities and then proceed to the linear perturbation equations.

RUT Perturbation Sprint - Step 4

FRW Background Reduction from the Minimal RUT Action

These results are labeled [D] where they follow from the action up to sign conventions. Open sign or coefficient issues are labeled [O].

FRW Setup [D]

Work in a flat FRW metric:

$$ds^2 = -dt^2 + a(t)^2 d\mathbf{x}^2$$

Take homogeneous fields:

$$\phi_{\text{bar}}(t), \chi_{\text{bar}}(t), \theta_{\text{bar}}(t)$$

Define the background bridge activation:

$$B_{\text{bar}}(t) = B(\chi_{\text{bar}}(t)) = 1/2 [1 + \tanh((\chi_{\text{bar}} - \chi_{\text{star}})/\Delta\chi)]$$

Assume the gauge-sector contribution ρ_A and p_A comes from an averaged electromagnetic or magneto-acoustic field with energy density and pressure $\rho_A(B)$ and $p_A(B)$.

Phase Field ϕ [D]

The scalar phase-order field background Lagrangian is:

$$\mathcal{L}_{\phi} = 1/2 K(B_{\text{bar}}) \dot{\phi}_{\text{bar}}^2 - V(\phi_{\text{bar}}, B_{\text{bar}})$$

It yields energy density and pressure:

$$\rho_{\phi} = 1/2 K(B_{\text{bar}}) \dot{\phi}_{\text{bar}}^2 + V(\phi_{\text{bar}}, B_{\text{bar}})$$

$$p_{\phi} = 1/2 K(B_{\text{bar}}) \dot{\phi}_{\text{bar}}^2 - V(\phi_{\text{bar}}, B_{\text{bar}})$$

Bridge Field χ [D]

The bridge field background Lagrangian is:

$$\mathcal{L}_{\chi} = 1/2 \dot{\chi}_{\text{bar}}^2 - U(\chi_{\text{bar}})$$

The corresponding energy density and pressure are:

$$\rho_{\chi} = 1/2 \dot{\chi}_{\text{bar}}^2 + U(\chi_{\text{bar}})$$

$$p_{\chi} = 1/2 \dot{\chi}_{\text{bar}}^2 - U(\chi_{\text{bar}})$$

Gauge Sector [D]

The gauge field contributes:

$$\mathcal{L}_A = -1/4 Z(B_{\text{bar}}) F_{\mu\nu} F^{\mu\nu}$$

In a homogeneous, statistically isotropic background one writes:

$$\rho_A = 1/2 Z(B_{\text{bar}}) \langle E^2 + B_{\text{em}}^2 \rangle$$

$$p_A = 1/6 Z(B_{\text{bar}}) \langle E^2 + B_{\text{em}}^2 \rangle$$

These reduce to zero if no background field is assumed.

Q-Sector [D/O]

For the Q-sector:

$$L_Q = -1/2 \, \xi_Q \, B_{\text{bar}} \, (\nabla^\mu \theta_{\text{bar}} \, \nabla_\mu \phi_{\text{bar}})^2$$

On FRW:

$$\nabla^\mu \theta_{\text{bar}} \, \nabla_\mu \phi_{\text{bar}} = - \, \dot{\theta}_{\text{bar}} \, \dot{\phi}_{\text{bar}}$$

Thus the invariant is:

$$X^2 == (\nabla^\mu \theta_{\text{bar}} \, \nabla_\mu \phi_{\text{bar}})^2 = \dot{\theta}_{\text{bar}}^2 \, \dot{\phi}_{\text{bar}}^2$$

The Lagrangian density is:

$$L_Q = -1/2 \, \xi_Q \, B_{\text{bar}} \, X^2$$

A full variation yields a stress-energy tensor of the schematic form:

$$T_{mn}(Q) = c_1 \, \xi_Q \, B_{\text{bar}} \, X \, (\partial_m \theta \, \partial_n \phi + \partial_m \phi \, \partial_n \theta) + c_2 \, \xi_Q \, B_{\text{bar}} \, X^2 \, g_{mn}$$

The coefficients c_1 and c_2 depend on the sign convention used. Selecting signs so the energy density is positive gives the schematic result:

$$\rho_Q \text{ proportional to } \xi_Q \, B_{\text{bar}} \, \dot{\theta}_{\text{bar}}^2 \, \dot{\phi}_{\text{bar}}^2$$

The exact prefactor, such as $5/2$ or $3/2$ depending on convention bookkeeping, must be fixed by a careful metric-variation sign check. This remains an open derivation target [O]. The pressure p_Q is likewise fixed by the full stress-energy tensor and needs the same consistent sign check.

Modified Friedmann Equations [D]

Collecting all sectors, the background Friedmann equation becomes:

$$3 \, M_{\text{Pl}}^2 \, H^2 = \rho_m + \rho_r + \rho_\phi + \rho_\chi + \rho_A + \rho_Q$$

The acceleration equation is:

$$-2 \, M_{\text{Pl}}^2 \, \dot{H} = \rho_{\text{tot}} + p_{\text{tot}}$$

with:

$$\rho_{\text{tot}} = \rho_m + \rho_r + \rho_\phi + \rho_\chi + \rho_A + \rho_Q$$

$$p_{\text{tot}} = p_m + p_r + p_\phi + p_\chi + p_A + p_Q$$

Remarks and Open Points

- The phase-field, bridge-field, and gauge contributions follow standard variational derivations and are fully determined by the action [D].
- The Q-sector contributes energy density and pressure quadratic in $\dot{\theta}_{\text{bar}} \dot{\phi}_{\text{bar}}$. Fixing the precise sign and coefficient requires a rigorous metric-variation sign check [O].
- In the limit $B_{\text{bar}} \rightarrow 0$, all bridge-dependent modifications vanish and the background reduces to GR plus minimally coupled scalar fields.

Next step: Step 5 sets up the linear perturbation equations.

RUT Perturbation Sprint - Step 5A

Bridge Perturbations and Coefficient Variations

Status label: [D] for definitions/derivatives that follow directly from the declared bridge profile; [W] for working perturbation setup; [O] for open items deferred to Step 5B and Step 6.

1. Purpose of Step 5A

Steps 1-4 fixed conventions, obtained schematic field equations, derived the Q-sector stress-energy structure, and reduced the model to a homogeneous FRW background. Step 5 begins the linear perturbation setup. This Step 5A document completes the bridge-side perturbations: the derivative of $B(\chi)$, the first-order bridge perturbation δB , and the induced variations of $K(B)$, $Z(B)$, and $V(\phi, B)$.

This is not yet the full linearized Einstein system. The remaining Step 5B task is to perturb $Q = \nabla^\mu \theta \nabla_\mu \phi$ and assemble the Q-sector perturbation sources, especially the off-diagonal spatial terms that can affect gravitational slip.

2. Bridge activation profile [D]

The bridge activation is defined as

$$B(\chi) = (1/2) [1 + \tanh((\chi - \chi_*) / \Delta \chi)].$$

Using $d(\tanh z)/dz = \text{sech}^2 z$, the derivative with respect to χ is

$$B_{,\chi} = (1 / (2 \Delta \chi)) \text{sech}^2((\chi - \chi_*) / \Delta \chi).$$

This derivative is sharply localized near the bridge transition χ approximately χ_* . Far from the transition, sech^2 becomes small and the bridge-induced perturbative corrections are suppressed.

3. Linear bridge perturbation [D/W]

Perturb the bridge field around its homogeneous background:

$$\chi(t, x) = \bar{\chi}(t) + \delta \chi(t, x).$$

Then expand $B(\chi)$ to first order:

$$\begin{aligned} B(t, x) &= \bar{B}(t) + \delta B(t, x), \\ \bar{B}(t) &= B(\bar{\chi}(t)), \\ \delta B &= B_{,\chi} |_{\bar{\chi}} \delta \chi. \end{aligned}$$

Therefore, explicitly,

$$\delta B = [1 / (2 \Delta \chi)] \text{sech}^2((\bar{\chi} - \chi_*) / \Delta \chi) \delta \chi.$$

Interpretation: the bridge perturbation is strongest when the background field is crossing the transition region. This gives RUT a built-in localization mechanism: bridge-driven modifications are naturally small away from the activation window.

4. Perturbations of $K(B)$, $Z(B)$, and $V(\phi, B)$ [D/W]

For any smooth bridge-modulated kinetic or gauge functions, the first-order variations are

$$\begin{aligned} \delta K &= K_{,B} \delta B, \\ \delta Z &= Z_{,B} \delta B. \end{aligned}$$

For the mixed potential $V(\phi, B)$, the first-order variation is

$$\delta V = V_{,\phi} \delta \phi + V_{,B} \delta B.$$

If the model choices are $K(B)=1+\xi_K B$ and $Z(B)=1+\xi_F B$, then the variations reduce to

$$\begin{aligned}\delta K &= \xi_K \delta B, \\ \delta Z &= \xi_F \delta B.\end{aligned}$$

For a representative potential

$$V(\phi, B) = \lambda (\phi^2 - \phi_0^2)^2 (1-B) + V_{\text{late}}(\phi) B,$$

one obtains the bridge derivative

$$V_{,B} = -\lambda (\phi^2 - \phi_0^2)^2 + V_{\text{late}}(\phi),$$

and the phase-field derivative

$$V_{,\phi} = 4 \lambda \phi (\phi^2 - \phi_0^2) (1-B) + B dV_{\text{late}}/d\phi.$$

These expressions show how a perturbation in χ propagates into the scalar kinetic term, the electromagnetic/gauge normalization, and the potential energy sector through δB .

5. Perturbation source list prepared for Step 6 [W]

The following first-order quantities are now available for insertion into the linearized Einstein and field equations:

- $\delta B = B_{,\chi} \delta \chi$
- $\delta K = K_{,B} \delta B$
- $\delta Z = Z_{,B} \delta B$
- $\delta V = V_{,\phi} \delta \phi + V_{,B} \delta B$
- δB is localized by $\text{sech}^2((\chi_{\text{bar}} - \chi_*)/\Delta \chi)$

6. What Step 5A does and does not complete

Completed in Step 5A:

- The derivative of the tanh bridge profile $B_{,\chi}$.
- The first-order bridge perturbation δB .
- The induced first-order variations δK , δZ , and δV .
- The interpretation that bridge perturbations are strongest near $\chi = \chi_*$ and suppressed away from the transition.

Still open for Step 5B:

- Derive δQ for $Q = \nabla^\mu \theta \nabla_\mu \phi$ in Newtonian gauge.
- Compute the Q-sector contributions to δT^0_0 , δT^0_i , and δT^i_j .
- Identify the off-diagonal spatial contribution that can source gravitational slip.
- Assemble the complete perturbation source list for Step 6 linearized Einstein equations.

7. Step 5B target expression for δQ [W/O]

The next calculation begins from

$$Q = g^{\mu\nu} \partial_\mu \theta \partial_\nu \phi.$$

At first order, the schematic perturbation is

$$\begin{aligned}\delta Q &= \delta g^{\mu\nu} \partial_\mu \theta_{\text{bar}} \partial_\nu \phi_{\text{bar}} \\ &+ g_{\text{bar}}^{\mu\nu} \partial_\mu \delta \theta \partial_\nu \phi_{\text{bar}} \\ &+ g_{\text{bar}}^{\mu\nu} \partial_\mu \theta_{\text{bar}} \partial_\nu \delta \phi.\end{aligned}$$

For homogeneous background fields in Newtonian gauge, this is expected to reduce schematically to

$$\begin{aligned} \delta Q &\sim 2 \Psi \bar{\theta}_{\dot{}} \bar{\phi}_{\dot{}} \\ &- \delta \theta_{\dot{}} \bar{\phi}_{\dot{}} \\ &- \bar{\theta}_{\dot{}} \delta \phi_{\dot{}}, \end{aligned}$$

with the exact signs fixed by the mostly-plus convention and the metric perturbation expansion. This is the first target for Step 5B.

Conclusion

Step 5A completes the bridge-side perturbation setup. The RUT bridge now has a clear first-order perturbation rule, and all bridge-modulated sectors inherit variations through δB . The next required derivation is Step 5B: perturb the Q-sector and identify the anisotropic-stress source that will feed the modified-gravity functions $\mu(k,a)$, $\Sigma(k,a)$, $\gamma(k,a)$, and the E_G statistic.

RUT Perturbation Sprint - Step 5B

Bridge and Coupling Perturbations

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Status label: This step completes the bridge/coupling perturbation setup needed before the full linearized Einstein and field equations. The results below are definitions and first-order expansions derived by chain rule from the bridge profile. The Q-sector perturbation, anisotropic-stress source terms, and the full linearized Einstein equations are reserved for Step 6.

Derivative of the Hyperbolic Tangent

The standard identity used here is

$$\frac{d}{dz} \tanh z = \text{sech}^2 z. \quad (1)$$

This identity is listed, for example, in Wolfram MathWorld's entry on the hyperbolic tangent.

Bridge Field and Its Perturbation

The bridge activation is defined by

$$B(\chi) = \frac{1}{2} \left[1 + \tanh \left(\frac{\chi - \chi_*}{\Delta\chi} \right) \right]. \quad (2)$$

Differentiating with respect to χ gives

$$B_{,\chi} = \frac{1}{2} \text{sech}^2 \left(\frac{\chi - \chi_*}{\Delta\chi} \right) \frac{1}{\Delta\chi} \quad (3)$$

$$= \frac{1}{2\Delta\chi} \text{sech}^2 \left(\frac{\chi - \chi_*}{\Delta\chi} \right). \quad (4)$$

At linear order, perturb the bridge field around its homogeneous background:

$$\chi(t, \mathbf{x}) = \bar{\chi}(t) + \delta\chi(t, \mathbf{x}). \quad (5)$$

The corresponding perturbation in the bridge activation is

$$\delta B = B_{,\chi} \delta\chi \quad (6)$$

$$= \frac{1}{2\Delta\chi} \text{sech}^2 \left(\frac{\bar{\chi} - \chi_*}{\Delta\chi} \right) \delta\chi. \quad (7)$$

Perturbations of Coupling Functions

Let $K(B)$, $Z(B)$, and $V(\phi, B)$ be smooth functions of their arguments. Their first-order variations are

$$\delta K = K_{,B} \delta B, \quad (8)$$

$$\delta Z = Z_{,B} \delta B, \quad (9)$$

$$\delta V = V_{,\phi} \delta \phi + V_{,B} \delta B. \quad (10)$$

Here $V_{,\phi} \equiv \partial V / \partial \phi$ and $V_{,B} \equiv \partial V / \partial B$, evaluated on the background.

Because $B_{,\chi}$ is proportional to

$$\text{sech}^2 \left(\frac{\bar{\chi} - \chi_*}{\Delta \chi} \right), \quad (11)$$

these perturbations are suppressed far from the bridge transition and become important near

$$\bar{\chi} \simeq \chi_*. \quad (12)$$

Core Step 5B Results

The essential first-order bridge/coupling perturbation package is therefore

$$B_{,\chi} = \frac{1}{2\Delta\chi} \text{sech}^2 \left(\frac{\bar{\chi} - \chi_*}{\Delta\chi} \right), \quad (13)$$

$$\delta B = B_{,\chi} \delta \chi, \quad (14)$$

$$\delta K = K_{,B} \delta B, \quad (15)$$

$$\delta Z = Z_{,B} \delta B, \quad (16)$$

$$\delta V = V_{,\phi} \delta \phi + V_{,B} \delta B. \quad (17)$$

Why This Matters

These relations supply the necessary bridge-side ingredients for the linearized Einstein and field equations. They show that the bridge contribution is localized by the sech^2 envelope: outside the transition region, the bridge perturbation is small; near χ_* , it becomes dynamically important.

What Moves to Step 6

Step 6 should begin by perturbing the Q-sector quantity

$$Q = \nabla^\mu \theta \nabla_\mu \phi, \quad (18)$$

so that

$$\delta Q = \delta (\nabla^\mu \theta \nabla_\mu \phi). \quad (19)$$

This is needed because the Q-sector can source anisotropic stress and therefore affect gravitational slip. Step 6 should then assemble the linearized 00, 0i, and ij Einstein equations, identify the sources of Φ and Ψ , and prepare the quasi-static reduction toward $\mu(k, a)$, $\Sigma(k, a)$, $\eta(k, a)$, and $E_G(k, a)$.

Conclusion

Step 5B completes the bridge and coupling perturbation setup. The result is not yet the full perturbation theory, but it provides the clean first-order inputs needed for the next derivation stage.

RUT Perturbation Sprint - Step 6

Linearized Einstein Equations, RUT Source Terms, and the Road to $\mu(k,a)$, $\Sigma(k,a)$, and $E_G(k,a)$

Status label: This step is a formal linearized-equations setup. The standard Einstein-equation structure is established, RUT source channels are identified, and the quasi-static definitions of μ and Σ are prepared. The exact Q-sector anisotropic-stress coefficient remains a next-step calculation.

1. Newtonian Gauge Setup

Use a flat FRW background with scalar perturbations in Newtonian, or longitudinal, gauge. To stay consistent with the previous convention sprint, this page uses cosmic time t and the mostly-plus metric signature.

$$ds^2 = -(1 + 2\Psi) dt^2 + a^2(t)(1 - 2\Phi) d\mathbf{x}^2$$

Here Ψ and Φ are the scalar metric potentials. The matter sector is described by density perturbation $\delta\rho$, pressure perturbation δp , momentum density q_i , and anisotropic stress π_{ij} .

2. Standard Linearized Einstein Equations

The conventional linearized equations supply the template into which the RUT source terms are inserted.

2.1 00-component: Hamiltonian constraint

$$\nabla^2\Phi - 3H(\dot{\Phi} + H\Psi) = 4\pi G \delta\rho$$

This equation becomes the modified Poisson relation once all RUT perturbative energy-density sources are included.

2.2 0i-component: momentum constraint

$$\partial_i(\dot{\Phi} + H\Psi) = 4\pi G q_i = 4\pi G(\bar{\rho} + \bar{p})v_i$$

This relates time variation of the potentials to the scalar velocity or momentum flow of the matter and field sectors.

2.3 ij trace: pressure equation

$$\ddot{\Phi} + H(\dot{\Psi} + 2\dot{\Phi}) + (2\dot{H} + 3H^2)\Psi - \frac{1}{3}\nabla^2(\Phi - \Psi) = 4\pi G \delta p$$

This equation tracks pressure and entropy-like perturbations. In RUT, it receives contributions from the phase, bridge, gauge, and Q sectors.

2.4 Traceless ij part: anisotropy constraint

$$\partial_i\partial_j(\Phi - \Psi) - \frac{1}{3}\delta_{ij}\nabla^2(\Phi - \Psi) = 8\pi G \pi_{ij}$$

This is the slip equation. Perfect fluids have $\pi_{ij} = 0$, giving $\Phi = \Psi$ in the standard GR limit. Any new anisotropic stress can make Φ differ from Ψ .

3. How RUT Modifies the Source Terms

RUT does not change the geometric form of the Einstein equations at this step. Instead, it adds new stress-energy sources from the fields in the action.

- **Phase field ϕ and bridge field χ .** Their perturbations enter the 00 and 0i equations through $\delta\rho_\phi$, $\delta\rho_\chi$, momentum-density terms, and pressure perturbations. The bridge-dependent functions $K(B)$ and $V(\phi, B)$ make these effects activate near the bridge transition.
- **Gauge sector.** The modulated Maxwell term $Z(B)F_{\{\mu\nu\}}F^{\{\mu\nu\}}$ contributes to energy density and pressure. In first-pass tests this sector may be averaged or treated as small, but it is the controlled route toward magneto-acoustic transport.
- **Q-sector.** The interaction term is the clearest RUT-specific source of possible anisotropic stress, because it couples gradients of θ and ϕ with the bridge prefactor $B(\chi)$.

4. Q-Sector Slip Source

From Step 3, the Q-sector stress-energy tensor is

$$T_{\mu\nu}^{(Q)} = \xi_Q B Q (\partial_\mu \theta \partial_\nu \phi + \partial_\mu \phi \partial_\nu \theta) - \frac{1}{2} \xi_Q B Q^2 g_{\mu\nu}$$

with

$$Q = \nabla^\alpha \theta \nabla_\alpha \phi$$

At the level of spatial off-diagonal components, the relevant structure is schematically

$$T_{ij}^{(Q)} \supset \xi_Q B Q (\partial_i \theta \partial_j \phi + \partial_i \phi \partial_j \theta) \quad (i \neq j)$$

In a strictly homogeneous background, spatial gradients vanish. At perturbation level, gradients of $\delta\theta$ and $\delta\phi$ can enter. Therefore, the traceless ij equation becomes the gatekeeper for RUT gravitational slip.

$$\partial_i \partial_j (\Phi - \Psi) = 8\pi G (\pi_{ij}^{\text{matter}} + \pi_{ij}^{(Q)} + \pi_{ij}^{(\Lambda)} + \dots)$$

Reviewer-safe caveat: A precise expression for $\pi_{ij}^{(Q)}$ requires expanding δQ , δB , and the spatial-gradient pieces of $\delta\phi$ and $\delta\theta$. This is the main technical task feeding Step 7.

5. Modified Poisson and Weyl Relations

Collecting all perturbative sources, the 00 equation becomes

$$-k^2 \Phi + 3H(\dot{\Phi} + H\Psi) = 4\pi G(\delta\rho_m + \delta\rho_\phi + \delta\rho_\chi + \delta\rho_Q + \delta\rho_\Lambda)$$

In the subhorizon quasi-static limit, time-derivative terms are suppressed and the relation motivates the effective modified-gravity function $\mu(k, a)$:

$$-k^2 \Psi = 4\pi G a^2 \mu(k, a) \bar{\rho}_m \delta_m$$

For lensing, define $\Sigma(k, a)$ through the Weyl potential:

$$-k^2 (\Phi + \Psi) = 8\pi G a^2 \Sigma(k, a) \bar{\rho}_m \delta_m$$

The GR limit is $\mu = \Sigma = 1$. In RUT, μ and Σ can differ once bridge-activated scalar, gauge, or Q-sector effects modify growth and lensing differently.

6. Step 6 Deliverables and Next Gate

- Linearized Einstein equations in Newtonian gauge are fixed as the template.
- RUT source terms are assigned to the 00, 0i, trace, and traceless equations.
- The Q-sector is identified as the cleanest possible source of gravitational slip through anisotropic stress.
- Operational definitions of $\mu(k,a)$ and $\Sigma(k,a)$ are prepared for the quasi-static reduction.
- The GR correspondence limit requires $B \rightarrow 0$, $\pi_{ij} \rightarrow 0$, $\Phi = \Psi$, $\mu = 1$, and $\Sigma = 1$.

Next step: Step 7 should expand δQ and the relevant pieces of $\delta T^{(Q)}_{ij}$, then take the quasi-static subhorizon limit to derive working forms for $\mu(k,a)$, $\Sigma(k,a)$, $\eta(k,a)$, and $E_G(k,a)$.

RUT Perturbation Sprint - Step 7A

Quasi-Static Reduction Roadmap, Revised

Status: Complete as a roadmap; not complete as a first-principles quasi-static derivation. The goal of this step is to state the subhorizon reduction program cleanly and preserve the bridge-activated working forms for later derivation.

1. Bridge derivative and first-order bridge perturbation

Using the identity $d(\tanh z)/dz = \text{sech}^2 z$, the derivative of the bridge activation function is:

$$B_{,\chi} = \frac{1}{2\Delta\chi} \text{sech}^2\left(\frac{\chi - \chi_*}{\Delta\chi}\right)$$

Therefore the first-order bridge perturbation is:

$$\delta B = B_{,\chi} \delta\chi$$

This enters all bridge-dependent quantities, including $K(B)$, $Z(B)$, and $V(\phi, B)$:

$$\delta K = K_{,B} \delta B, \quad \delta Z = Z_{,B} \delta B, \quad \delta V = V_{,\phi} \delta\phi + V_{,B} \delta B$$

2. Linearized Einstein-equation target

In Newtonian gauge, the linearized Einstein equations involve the scalar potentials Φ and Ψ . The RUT fields add perturbations from the phase field, bridge field, light-phase field, gauge sector, and Q-sector.

The Q-sector is especially important because it can source anisotropic stress through cross-gradient terms such as:

$$\partial_i \bar{\theta} \partial_j (\delta\phi) + \partial_i \bar{\phi} \partial_j (\delta\theta)$$

Those terms can source $\Phi - \Psi$ and therefore affect gravitational slip. This is the bridge between the action-level Q-term and the observational functions $\mu(k, a)$, $\Sigma(k, a)$, and $E_G(k, a)$.

3. Quasi-static working ansatz

Guided by modified-gravity quasi-static reductions, the expected bridge-activated forms are:

$$\mu(k, a) \approx 1 + \frac{\xi_\mu B(a) k^2}{k^2 + a^2 m_\phi^2}$$

$$\Sigma(k, a) \approx 1 + \frac{\xi_\Sigma B(a) k^2}{k^2 + a^2 m_\Sigma^2}$$

Here ξ_μ , ξ_Σ , m_ϕ , and m_Σ should eventually be expressed in terms of action parameters once the algebra is carried out. In this Step 7A version, they remain working parameters.

These forms recover the GR limit when the bridge is inactive, $B \rightarrow 0$, or when the scalar fields are sufficiently massive.

4. Remaining derivation: Step 7B

- Take the quasi-static subhorizon limit $k \gg aH$, neglecting time derivatives of perturbations relative to spatial gradients where justified.
- Solve the algebraic perturbation equations for $\delta\phi$, $\delta\chi$, and $\delta\theta$ in terms of the matter overdensity δ_m and/or the metric potentials.
- Substitute those solutions back into the linearized Einstein equations to read off the modified Poisson and Weyl relations.
- Derive explicit expressions for $\mu(k,a)$ and $\Sigma(k,a)$, identifying which coefficients are genuine combinations of $K(B)$, V_ϕ , U_χ , B_χ , $Z(B)$, and ξ_Q .
- Verify the GR limit $B \rightarrow 0$, $\mu \rightarrow 1$, and $\Sigma \rightarrow 1$, then check the same parameter set against background, growth, and lensing data.

5. Clean downstream definitions

Once $\mu(k,a)$ and $\Sigma(k,a)$ are obtained, the slip and E_G statistic are defined by:

$$\eta(k, a) = 2 \frac{\Sigma(k, a)}{\mu(k, a)} - 1$$

$$E_G(k, a) = \frac{\Omega_{m,0} \Sigma(k, a)}{f(a)}$$

This convention keeps the perturbation program reviewer-safe: μ modifies the growth potential, Σ modifies the Weyl/lensing potential, η is derived from μ and Σ , and E_G becomes the growth-lensing falsifier.

6. Summary label

Step 7A is complete as a roadmap. It turns the quasi-static stage from a general discussion into a clear derivational program. Step 7B remains open: explicitly solve the subhorizon perturbation equations and derive $\mu(k,a)$, $\Sigma(k,a)$, $\eta(k,a)$, and $E_G(k,a)$ from the RUT action.

RUT Perturbation Sprint - Step 7B

Quasi-Static Matrix Reduction and Structural Forms of $\mu(k,a)$ and $\Sigma(k,a)$

Status: structurally derived / matrix-level reduction. Exact coefficients remain open until the full quadratic perturbation action is computed.

Purpose

Step 7B upgrades the earlier Yukawa-like working ansatz into a reviewer-safe structural derivation. The key point is that, in local modified-gravity theories, quasi-static response functions commonly appear as ratios of even polynomials in k . RUT obtains the same pattern because the perturbations are solved by inverting a mass-gradient matrix whose diagonal entries contain k^2/a^2 plus effective masses.

The quasi-static approximation must be restricted to subhorizon and sub-sound-horizon scales. RUT should not apply these equations to ISW-scale, horizon-scale, or sound-horizon-scale claims without a full numerical perturbation treatment.

$$\frac{k}{a} \gg H, \quad \frac{k}{a} \gg m_{\text{sound/horizon}}^{-1}$$

1. Perturbation variables and bridge response

$$\phi = \bar{\phi} + \delta\phi, \quad \chi = \bar{\chi} + \delta\chi, \quad \theta = \bar{\theta} + \delta\theta$$

$$\delta B = B_{,\chi} \delta\chi, \quad B_{,\chi} = \frac{1}{2\Delta\chi} \text{sech}^2\left(\frac{\bar{\chi} - \chi^*}{\Delta\chi}\right)$$

$$\delta K = K_{,B} B_{,\chi} \delta\chi, \quad \delta Z = Z_{,B} B_{,\chi} \delta\chi, \quad \delta V = V_{,\phi} \delta\phi + V_{,B} B_{,\chi} \delta\chi$$

These bridge and coefficient variations are derived from the bridge definition and the chain rule. The bridge response is localized near the transition because $B_{,\chi}$ is proportional to sech squared.

2. Quasi-static scalar perturbation system

In the subhorizon limit $k^2/a^2 \gg H^2$, spatial gradients dominate over time derivatives for the relevant scalar perturbations. The coupled perturbation equations become approximately algebraic:

$$\left(\frac{k^2}{a^2} + m_\phi^2\right)\delta\phi + m_{\phi\chi}^2\delta\chi + m_{\phi\theta}^2\delta\theta \simeq S_\phi(\Psi, \Phi, \delta_m)$$

$$m_{\chi\phi}^2\delta\phi + \left(\frac{k^2}{a^2} + m_\chi^2\right)\delta\chi + m_{\chi\theta}^2\delta\theta \simeq S_\chi(\Psi, \Phi, \delta_m)$$

$$m_{\theta\phi}^2\delta\phi + m_{\theta\chi}^2\delta\chi + \left(c_\theta^2 \frac{k^2}{a^2} + m_\theta^2\right)\delta\theta \simeq S_\theta(\Psi, \Phi, \delta_m)$$

Label: these equations are [M]/[O] - a quasi-static reduced structure, not yet a fully solved system with every coefficient derived.

3. Effective masses and mixing terms

The effective masses are action-level combinations. Schematic examples are:

$$m_\phi^2 \sim \frac{1}{K(\bar{B})} V_{,\phi\phi} + \dots$$

$$m_\chi^2 \sim U_{,\chi\chi} + B_{,\chi\chi} \left[\frac{1}{2} K_{,B} \dot{\bar{\phi}}^2 - V_{,B} + \frac{1}{2} \xi_Q \bar{Q}^2 \right] + B_{,\chi}^2 \left[\frac{1}{2} K_{,BB} \dot{\bar{\phi}}^2 - V_{,BB} + \frac{1}{2} (\xi_Q)_{,B} \bar{Q}^2 \right]$$

$$\bar{Q} = \nabla^\mu \bar{\theta} \nabla_\mu \bar{\phi} = -\dot{\bar{\theta}} \dot{\bar{\phi}}$$

$$m_{\phi\chi}^2 \sim \frac{1}{K(\bar{B})} \left[V_{,\phi B} B_{,\chi} - K_{,B} B_{,\chi} (\nabla^2 \bar{\phi}) + \dots \right]$$

These expressions identify the correct action-level origins of the masses and mixings. Final numerical coefficients require a sign-checked quadratic action.

4. Matrix solution structure

$$\mathbf{M}(k, a) \delta \boldsymbol{\varphi} = \mathbf{S}$$

$\mathbf{M}(k, a)$ has diagonal entries $k^2/a^2 + m_{\text{phi}}^2$, $k^2/a^2 + m_{\text{chi}}^2$, and $c_{\text{theta}}^2 k^2/a^2 + m_{\text{theta}}^2$, with off-diagonal mixing entries $m_{\text{phi_chi}}^2$, $m_{\text{phi_theta}}^2$, $m_{\text{chi_theta}}^2$, and their transposed partners.

$$\delta \boldsymbol{\varphi} = \mathbf{M}^{-1}(k, a) \mathbf{S}$$

$$\delta \varphi_A = [\mathbf{M}^{-1}(k, a)]_{AB} S_B, \quad \varphi_A = (\phi, \chi, \theta)$$

Equation status: [D/M]. This is derived as a quasi-static algebraic reduction, while the exact matrix entries remain open.

5. Structural origin of $\mu(k, a)$

The modified Poisson equation is written as:

$$-k^2 \Psi = 4\pi G a^2 \bar{\rho}_m \delta_m + a^2 \delta \rho_{\text{RUT}}^{\text{QS}}$$

$$\delta \rho_{\text{RUT}}^{\text{QS}} = C_A [\mathbf{M}^{-1}(k, a)]_{AB} S_B$$

If the dominant source is proportional to the matter overdensity, $S_B \approx \alpha_B(a) \rho_m \delta_m$, then:

$$\mu(k, a) = 1 + \frac{C_A [\mathbf{M}^{-1}(k, a)]_{AB} \alpha_B(a)}{4\pi G}$$

Equation status: [D/M]. This is stronger than the earlier ansatz because it shows where μ comes from: the inverse perturbation mass matrix.

6. Reduced one-field approximation for mu

If delta phi dominates and delta chi, delta theta mixing is subleading, then:

$$\delta\phi \simeq \frac{\alpha_\phi(a)\bar{\rho}_m\delta_m}{k^2/a^2 + m_\phi^2}$$

$$\mu(k, a) \approx 1 + \frac{\xi_\mu(a)B(a)k^2}{k^2 + a^2m_\phi^2}$$

Equation status: [A/M]. This is the effective one-channel test function. It is not a fully derived prediction until xi_mu(a) is computed from the full action.

7. Structural origin of Sigma(k,a)

The Weyl/lensing equation is written as:

$$-k^2(\Phi + \Psi) = 8\pi G a^2 \bar{\rho}_m \delta_m + a^2 \delta\rho_{\text{Weyl}}^{\text{RUT}} + a^2 \Pi_{\text{RUT}}$$

$$\delta\rho_{\text{Weyl}}^{\text{RUT}} + \Pi_{\text{RUT}} = D_A [\mathbf{M}^{-1}(k, a)]_{AB} \alpha_B(a) \bar{\rho}_m \delta_m$$

$$\Sigma(k, a) = 1 + \frac{D_A [\mathbf{M}^{-1}(k, a)]_{AB} \alpha_B(a)}{8\pi G}$$

Equation status: [D/M]. The structure is derived; the coefficients remain open.

$$\Sigma(k,a) \approx 1 + \frac{\xi_\Sigma(a)B(a)k^2}{k^2 + a^2 m_\Sigma^2}$$

Equation status: [A/M]. This is the effective one-channel Weyl/lensing test function.

8. Q-sector contribution and slip risk

$$L_Q = -\frac{1}{2}\xi_Q B(\chi) Q^2, \qquad Q = \nabla^\mu \theta \nabla_\mu \phi$$

$$\delta Q = \delta g^{\mu\nu} \partial_\mu \bar{\theta} \partial_\nu \bar{\phi} + \bar{g}^{\mu\nu} \partial_\mu \delta \theta \partial_\nu \bar{\phi} + \bar{g}^{\mu\nu} \partial_\mu \bar{\theta} \partial_\nu \delta \phi$$

$$\delta Q \sim 2\Psi \dot{\bar{\theta}} \dot{\bar{\phi}} - \delta \dot{\theta} \dot{\bar{\phi}} - \dot{\bar{\theta}} \delta \dot{\phi}$$

In strict quasi-static reduction one often drops time derivatives of perturbations, giving δQ_{QS} roughly proportional to $2\Psi \dot{\theta} \dot{\phi}$. But this is delicate. If θ carries spatial gradients or propagating modes, dropping time derivatives may erase important anisotropic stress. Therefore the Q-sector is the leading candidate source for $\Phi \neq \Psi$, but its exact quasi-static contribution remains open.

9. Final Step 7B result and status labels

The strongest reviewer-safe result is the matrix-response structure:

$$\mu(k,a) = 1 + \frac{C_A [\mathbf{M}^{-1}(k,a)]_{AB} \alpha_B(a)}{4\pi G}$$

$$\Sigma(k, a) = 1 + \frac{D_A [\mathbf{M}^{-1}(k, a)]_{AB} \alpha_B(a)}{8\pi G}$$

Under a reduced one-field or effective-one-channel approximation:

$$\mu(k, a) \approx 1 + \frac{\xi_\mu B(a) k^2}{k^2 + a^2 m_\phi^2}, \quad \Sigma(k, a) \approx 1 + \frac{\xi_\Sigma B(a) k^2}{k^2 + a^2 m_\Sigma^2}$$

Derived / structurally derived [D/M]: the inverse-matrix response for perturbations, μ , and Σ .

Working parameterizations [A/M]: the reduced Yukawa-like bridge-activated forms for μ and Σ .

Still open [O]: exact ξ_μ , ξ_Σ , m_ϕ , m_Σ , and the full Q-sector anisotropic-stress tensor from the sign-checked quadratic action.

10. GR correspondence limit

If $B(a) \rightarrow 0$, the bridge shuts off: $\delta B \rightarrow 0$, $\delta K \rightarrow 0$, $\delta Z \rightarrow 0$, and $L_Q \rightarrow 0$. Therefore:

$$\mu(k, a) \rightarrow 1, \quad \Sigma(k, a) \rightarrow 1, \quad \eta(k, a) \rightarrow 1, \quad E_G(k, a) \rightarrow \frac{\Omega_{m,0}}{f(a)}$$

Bottom line: Step 7B is partially completed at the structural derivation level. RUT has moved from guessed Yukawa-like forms to inverse perturbation-mass-matrix responses. The next true math-closure target is to compute the exact entries of $M(k, a)$, C_A , D_A , and α_A from the quadratic action.

RUT Perturbation Sprint - Step 7C

Quadratic Action and Coefficient Extraction

This step gives a reviewer-safe quadratic-action scaffold for extracting the coefficient structure behind the quasi-static functions $\mu(k,a)$ and $\Sigma(k,a)$. Exact numerical coefficients remain open unless the full second-order expansion is performed term by term. The central upgrade is that the earlier working forms are now connected to a perturbation mass/mixing matrix rather than being free guesses.

Status label

[D/M] Complete at the schematic quadratic-action level. [0] Exact coefficient extraction from the full second-order action remains open.

1. Quadratic expansion of the action

Perturb the fields and metric around a flat FRW background:

$\phi = \bar{\phi} + \delta\phi$, $\chi = \bar{\chi} + \delta\chi$, $\theta = \bar{\theta} + \delta\theta$ $ds^2 = -(1+2\Psi)dt^2 + a(t)^2 (1-2\Phi) d\mathbf{x}^2$

After inserting these perturbations into the minimal RUT action and expanding to second order, the scalar perturbation sector can be written schematically as:

$S^{(2)} = \frac{1}{2} \int d^3x dt a^3 [(\delta\phi_{,i})^T K_{AB}(\bar{B}) (\delta\phi_{,i}) - (k^2/a^2) (\delta\phi)^T G_{AB}(\bar{B}) (\delta\phi) - (\delta\phi_{,i})^T M_{AB}^2(\bar{B}) (\delta\phi_{,i}) + \text{metric-field mixing} + \text{metric-metric terms}]$ $\delta\phi_A = (\delta\phi, \delta\chi, \delta\theta)$

The metric potentials Φ and Ψ are then removed using the Einstein constraint equations when one wants the reduced scalar-sector dynamics. This produces the matrix system used in Step 7B.

2. Kinetic, gradient, and mass/mixing matrices

The quadratic system separates into a kinetic matrix, a gradient matrix, and a mass/mixing matrix. In the simplest background, the diagonal entries of the kinetic and gradient matrices are:

$K_{\phi\phi} = K(\bar{B})$, $K_{\chi\chi} = 1$, $K_{\theta\theta} = \xi_Q \bar{B} \dot{\bar{\phi}}^2$ $G_{\phi\phi} = K(\bar{B})$, $G_{\chi\chi} = 1$, $G_{\theta\theta} = \xi_Q \bar{B} \dot{\bar{\phi}}^2$

Off-diagonal kinetic and gradient mixings arise from the Q-sector coupling. The full quasi-static matrix is:

$M_{AB}(k,a) = (k^2/a^2) G_{AB} + M_{AB}^2(a)$

The mass/mixing matrix has the schematic form:

$M_{AB}^2 = [M_{\phi\phi}^2 \ M_{\phi\chi}^2 \ M_{\phi\theta}^2 \ M_{\chi\phi}^2 \ M_{\chi\chi}^2 \ M_{\chi\theta}^2 \ M_{\theta\phi}^2 \ M_{\theta\chi}^2 \ M_{\theta\theta}^2]$

Representative entries

The following entries identify where the action-level derivatives enter. They are not final coefficient-level results; they are schematic entries showing the correct source locations.

$M_{\phi\phi}^2 \sim V_{\phi\phi} / K(\bar{B}) + [K_{,B}/K] \bar{B} \dot{\bar{\phi}} + \text{Q-sector mass terms}$
 $M_{\chi\chi}^2 \sim U_{\chi\chi} + B_{\chi\chi} [-1/2 K_{,B} \dot{\bar{\phi}}^2 - V_{,B} - 1/4 Z_{,B} \langle F^2 \rangle + \text{Q-coupling terms}] + (B_{\chi})^2 [K_{,BB} \dot{\bar{\phi}}^2 + \dots]$
 $M_{\theta\theta}^2 \sim \xi_Q \bar{B} \dot{\bar{\phi}}^2 m_{\theta}^2$ (minimal angular θ case: $m_{\theta} = 0$ unless generated by coupling/background terms)
 $M_{\phi\chi}^2 \sim [V_{\phi} B / K(\bar{B})] B_{\chi} + [K_{,B} / K(\bar{B})] B_{\chi} \ddot{\bar{\phi}} + \text{Q-sector mixing}$
 $M_{\phi\theta}^2$ and $M_{\chi\theta}^2$ arise from expanding: $L_Q = -1/2 \xi_Q B(\chi) (\nabla\theta \cdot \nabla\phi)^2$

The Q-sector also produces metric-field mixing through Ψ and Φ . After the constraints are solved, those terms contribute to effective mass-like and source terms in the reduced scalar system.

3. Where different derivatives enter

Derivative / coupling	Where it enters	Status
K_B and K_{BB}	Kinetic prefactor, bridge-induced mass corrections, and terms proportional to $(\nabla \phi)^2$ and δB .	[D/M]
V_{phi phi}, V_{phi B}, V_{BB}	Bare phase-field mass, phi-chi mixing via B _{chi} , and bridge-sector curvature corrections.	[D/M]
U_{chi chi}	Bare mass of the bridge/order-parameter field chi.	[D]
B_{chi} and B_{chi chi}	Appear because K, Z, and V depend on B(chi); they localize bridge effects near the transition.	[D]
Z_B	Gauge-sector corrections to chi mass/mixing when background electromagnetic or magneto-acoustic energy is present.	[M/O]
xi_Q	Kinetic/gradient mixing, Q-sector mass-like entries, and anisotropic-stress sources.	[D/M]

4. Source vectors for observables

To connect the quadratic action to observables, define source vectors that appear after combining the field perturbation equations with the Einstein equations:

Vector	Role	Status
C_A	Determines how δvarphi_A contributes to the modified Poisson equation through the perturbed RUT energy density.	[M/O]
D_A	Determines how δvarphi_A contributes to the Weyl/lensing potential, including pressure and anisotropic stress.	[M/O]
alpha_A	Describes how matter density perturbations source the scalar perturbations after constraints are solved.	[M/O]

With these vectors, the structural modified-gravity functions are:

$$\mu(k,a) = 1 + [C_A (M^{-1})^{(AB)} \alpha_B] / (4 \pi G) \quad \Sigma(k,a) = 1 + [D_A (M^{-1})^{(AB)} \alpha_B] / (8 \pi G)$$

These equations show how the inverse mass/mixing matrix controls the scale-dependent modifications to the Poisson and Weyl equations.

5. Status audit

Item	Status	Explanation
Quadratic expansion structure	[D/M]	Follows from expanding the action to second order, but exact entries require term-by-term expansion.
M_{AB}(k,a) = (k²/a²)G_{AB} + M²_{AB}	[D/M]	The separation into gradient and mass/mixing terms is structurally derived.
Individual entries M_{phi phi}², M_{chi chi}², etc.	[M]	Sources are identified; exact numerical coefficients remain open.
Source vectors C_A, D_A, alpha_A	[M/O]	Roles are stated but explicit coefficients require the constraint equations.
mu and Sigma in terms of M⁻¹	[D/M]	Structurally derived from the matrix reduction; numerical implementation awaits full coefficient extraction.

6. Summary

Step 7C shows how the RUT action leads, after quadratic expansion, to a coupled system of scalar perturbations described by a kinetic matrix, a gradient matrix, and a mass/mixing matrix. The crucial insight is that the inverse of this matrix controls the scale-dependent modifications to the Poisson and Weyl equations. This explains why quasi-static functions such as $\mu(k,a)$ and $\Sigma(k,a)$ naturally take rational forms in k^2 .

The next open target is exact coefficient extraction: compute every entry of M²_{AB} and the source vectors C_A, D_A, and alpha_A from the full sign-checked quadratic action.

RUT Perturbation Sprint - Step 8

Gravitational Slip from Reviewer-Safe Definitions

1. Definitions recap [I]

We define the modified-gravity functions in Newtonian gauge by their operational roles in matter growth and lensing:

$$- k^2 \Psi = 4 \pi G a^2 \mu(k, a) \rho_{m_bar} \delta_m$$

$$- k^2 (\Phi + \Psi) = 8 \pi G a^2 \Sigma(k, a) \rho_{m_bar} \delta_m$$

Here ρ_{m_bar} is the background matter density and δ_m is the matter overdensity. In General Relativity, $\mu = \Sigma = 1$. These are definitions at this stage, not yet RUT-specific derivations.

2. Deriving the slip relation [D]

Define the gravitational slip as the potential ratio:

$$\eta(k, a) = \Phi / \Psi$$

Write the Weyl potential in terms of Ψ :

$$\Phi + \Psi = \Psi (1 + \Phi/\Psi)$$

Then:

$$- k^2 (\Phi + \Psi) = - k^2 \Psi (1 + \Phi/\Psi)$$

Using the modified Poisson definition:

$$- k^2 \Psi = 4 \pi G a^2 \mu \rho_{m_bar} \delta_m$$

we get:

$$- k^2 (\Phi + \Psi) = 4 \pi G a^2 \mu \rho_{m_bar} \delta_m (1 + \Phi/\Psi)$$

Equating this with the Weyl definition:

$$4 \pi G a^2 \mu \rho_{m_bar} \delta_m (1 + \Phi/\Psi) = 8 \pi G a^2 \Sigma \rho_{m_bar} \delta_m$$

Canceling common factors gives:

$$\mu (1 + \Phi/\Psi) = 2 \Sigma$$

Therefore:

$$\eta(k, a) = \Phi/\Psi = 2 \Sigma(k, a)/\mu(k, a) - 1$$

This result is derived directly from the definitions of μ and Σ . It does not require choosing a RUT-specific model for either function yet.

3. GR limit [D]

If the bridge is inactive, $B \rightarrow 0$, or if the extra scalar-sector response is suppressed, then the modified-gravity functions return to their GR values:

$$\mu \rightarrow 1, \text{ Sigma} \rightarrow 1$$

The slip relation then gives:

$$\eta(k, a) = 2(1/1) - 1 = 1$$

Thus $\Phi = \Psi$, matching the standard GR result for vanishing anisotropic stress in this convention.

4. Status summary

- The slip relation $\eta = 2 \text{ Sigma} / \mu - 1$ is derived from definitions and is therefore labeled [D].
- The specific functional forms of $\mu(k, a)$ and $\text{Sigma}(k, a)$ remain working ansatzes from Step 7A until Step 7B performs the explicit quasi-static reduction.
- This convention keeps the GR limit clean: $\mu = \text{Sigma} = 1$ implies $\eta = 1$.
- Step 9 can now define E_G using Sigma and $f(a)$, while Step 7B remains the open derivation target for μ and Sigma from the RUT action.

RUT Perturbation Sprint - Step 9

The E_G Falsifier

This step uses the gravitational slip relation from Step 8 and the reviewer-safe definitions of $\mu(k,a)$ and $\Sigma(k,a)$ to obtain a clean expression for the E_G statistic. The result is labeled carefully: the E_G definition and slip relation are derived from the declared conventions, while the current forms of μ and Σ remain working ansatzes until Step 7B is completed.

1. Recap of Definitions

From Step 6, the modified Poisson and lensing equations were defined as:

$$\begin{aligned} -k^2 \Psi &= 4\pi G a^2 \mu(k,a) \bar{\rho}_m \delta_m \\ -k^2(\Phi + \Psi) &= 8\pi G a^2 \Sigma(k,a) \bar{\rho}_m \delta_m \end{aligned}$$

In Step 8, the gravitational slip was derived as:

$$\eta(k,a) \equiv \Phi / \Psi = 2 \Sigma(k,a) / \mu(k,a) - 1.$$

This reproduces the GR limit $\eta = 1$ when $\mu = \Sigma = 1$.

2. Definition of the E_G Statistic

The E_G statistic combines galaxy-CMB lensing correlations and redshift-space distortions to test modified gravity. For a scale k , or equivalently for a multipole l at a redshift/redshift bin, define:

$$E_G(k,a) = \Omega_{m,0} \Sigma(k,a) / f(k,a).$$

Here $\Omega_{m,0}$ is the present-day matter fraction, and $f(k,a)$ is the linear growth rate:

$$f(k,a) = d \ln D(k,a) / d \ln a.$$

The growth factor $D(k,a)$ satisfies the modified growth equation:

$$D'' + [2 + H'/H] D' - (3/2) \Omega_m(a) \mu(k,a) D = 0,$$

with primes denoting derivatives with respect to $\ln a$. The appearance of $\mu(k,a)$ in this equation shows how modifications to the effective Newtonian strength affect growth.

3. RUT Prediction

Using the quasi-static RUT ansatz from Step 7A:

$$\begin{aligned} \mu(k,a) &\approx 1 + [\xi_\mu B(a) k^2] / [k^2 + a^2 m_\phi^2], \\ \Sigma(k,a) &\approx 1 + [\xi_\Sigma B(a) k^2] / [k^2 + a^2 m_\Sigma^2]. \end{aligned}$$

The E_G statistic becomes:

$$E_G(k,a) \approx [\Omega_{m,0} / f(k,a)] \times [1 + \xi_\Sigma B(a) k^2 / (k^2 + a^2 m_\Sigma^2)].$$

Because $f(k,a)$ itself is suppressed or enhanced by $\mu(k,a)$ through the growth equation, the RUT prediction for E_G is generally scale- and time-dependent.

GR limit

When the bridge is inactive, $B \rightarrow 0$, or the scalar masses m_ϕ and m_Σ are very large, the model recovers:

$$\begin{aligned} \mu &\rightarrow 1, \Sigma \rightarrow 1, \eta \rightarrow 1, \\ E_G(k,a) &\rightarrow \Omega_{m,0} / f(a). \end{aligned}$$

This reproduces the GR result in the declared convention.

Bridge-active regime

When $B(a)$ grows toward unity, Σ can depart from unity and E_G can deviate from $\Omega_m/2$. The transition scale is set by $k \approx a m_\Sigma$. For $k^2 \gg a^2 m_\Sigma^2$, the modification saturates at:

$$\Sigma(k,a) \approx 1 + \xi_\Sigma B(a).$$

4. Comments on Derivation Status

- The expression for E_G is derived from the definitions of μ , Σ , and the growth-rate relation used in this convention. [D/I]
- The slip relation $\eta = 2\Sigma/\mu - 1$ is derived from the Poisson and Weyl definitions. [D]
- The particular bridge-activated forms of μ and Σ are still working ansatzes from Step 7A. [A/M]
- Once $\mu(k,a)$ and $\Sigma(k,a)$ are derived in Step 7B from the coupled perturbation equations, the E_G expression becomes a firm falsifiable prediction.

5. Next Steps

To complete Step 7B and fully specify ξ_μ , ξ_Σ , m_ϕ , and m_Σ from the RUT action, the next derivation must solve the linear perturbation equations for $\delta\phi$, $\delta\chi$, and $\delta\theta$ in the sub-horizon limit, express them in terms of δ_m , and substitute those solutions back into the Einstein equations.

After that, Step 10 should audit the free parameters and Step 11 should introduce at least one numerical forecast for E_G versus redshift and scale.

Compact status label: Step 9 gives a clean E_G falsifier formula using the reviewer-safe μ and Σ convention. The formula is ready for simulations, but its coefficients remain provisional until Step 7B derives μ and Σ from the full perturbation system.

RUT Perturbation Sprint - Step 10

Parameter Audit and Minimal First-Paper Model Discipline

Purpose

Step 10 protects the first RUT perturbation paper from the most serious reviewer criticism: that the model could fit data only because it contains too many flexible parameters. The goal is to separate fundamental action parameters, derived effective parameters, and deferred or dangerous parameters, then define the smallest testable model for a first paper.

Core reviewer-safe rule:

Use the smallest bridge-activated μ/Σ model first, and defer the full magneto-acoustic, geometry-factor, and ESZI-inversion sectors until they are derived or independently constrained.

Recommended Minimal First-Paper Strategy

The first perturbation paper should focus on the bridge-activated growth and lensing sector. It can use effective phenomenological test parameters for $\mu(k,a)$ and $\Sigma(k,a)$, while clearly stating that these are placeholders for combinations of the action-level parameters and the quasi-static mass-matrix response.

Recommended first-pass test functions:

$$\mu(k,a) \sim 1 + [x_{\mu}(a) B(a) k^2] / [k^2 + a^2 m_{\phi}^2]$$
$$\Sigma(k,a) \sim 1 + [x_{\Sigma}(a) B(a) k^2] / [k^2 + a^2 m_{\Sigma}^2]$$

These functions should be labeled as working parameterizations until x_{μ} , x_{Σ} , m_{ϕ} , and m_{Σ} are computed from the full quadratic perturbation action.

Derived Effective Parameters

Parameter	Role	Origin	Fit status	Overfitting risk
m_{ϕ}	Effective mass of the phase field ϕ ; enters the quasi-static denominator $k^2 + a^2 m_{\phi}^2$. Approximate origin: $m_{\phi}^2 \sim V_{,\phi\phi} / K$.	Derived from action-level parameters such as λ , ϕ_0 , x_K , and the background bridge value B .	Do not treat as fundamental. Compute from $V_{,\phi\phi}$ and $K(B)$ when possible; otherwise use as a controlled nuisance parameter.	Moderate if independent; low if derived.
m_{χ}	Effective mass of the bridge field χ ; controls how bridge perturbations propagate and whether χ can be treated as heavy/frozen.	Derived from $U(\chi)$, $B_{,\chi}$, $B_{,\chi\chi}$, and bridge-coupling corrections.	Not needed as an independent first-paper fit parameter if χ is taken heavy/static.	Low if heavy/frozen; moderate if free.
m_{Σ}	Effective mass scale controlling the lensing modification Σ ; arises from the coupled perturbation matrix and anisotropic-stress sector.	Derived from the full mass matrix $M(k,a)$, including χ , θ , gauge, and Q-sector contributions.	Do not treat as fundamental. Use only as an effective nuisance parameter until matrix entries are computed.	Moderate if introduced as free.
x_{μ} , x_{Σ}	Effective couplings in the bridge-activated quasi-static forms for μ and Σ . They encode strength of growth and lensing modifications.	Should be derived from x_Q , x_K , x_F , masses, background fields, and the inverse perturbation mass matrix.	May be fitted initially as working parameters, but label as effective combinations, not fundamental constants.	Moderate-high if fitted separately; must eventually tie to action parameters.
$\alpha_B(a)$, C_A , D_A	Coefficients appearing in the source vectors and matrix-inversion forms for μ and Σ , as in Step 7B equations 7B.5-7B.7.	Derived from the perturbation equations and stress-energy response.	Do not introduce independently in the likelihood. Absorb into x_{μ} and x_{Σ} for the first paper.	Low if derived; high if separately fitted.

Deferred or Dangerous Parameters

Parameter	Role	Origin	Fit status	Overfitting risk
g	Geometry factor scaling the oscillation center frequency ω ; sometimes suggested as $\pi/\phi/e/\sqrt{n}$ or an $I_{0\text{inf}}$ mapping.	Interpretive construct; no action-level derivation yet.	Exclude from first physics paper. Keep in Master book or future theory note.	High: unconstrained and uninterpretable in empirical fit.
$I_{0\text{inf}}$	Zero-infinity or ESZI inversion operator mapping inner and outer boundary states.	Not specified as a symmetry of the present action.	Exclude from first physics paper. Reserve for Master version and future symmetry work.	High if included without derivation.
β_{-} , β_{+} , β_x	Weights of slow/fast magnetosonic variance and cross-correlation in the enlarged horizon sector.	Currently phenomenological scan weights; no primordial MHD derivation yet.	Suppress explicit magneto-acoustic backreaction in first perturbation paper by setting these to benchmark-zero or excluding them from the likelihood model.	High: multiple flexible parameters.
Other late-time or large-scale bridge factors	Examples: light-time inversion scale, extra phase transitions, additional geometry-light coupling terms.	Interpretive or speculative; not part of the minimal first-paper action.	Exclude from first paper unless mathematically forced by the action.	High: not needed for core model.
ξ_Q amplitude	Q-sector coupling controlling light-phase/phase-field mixing and possible anisotropic stress.	Present in the action, but sign, amplitude, and perturbative stability still require quadratic-action checking.	For a first empirical paper, fix ξ_Q to zero or a small benchmark value unless the Q-sector is explicitly analyzed.	Moderate-high if left free.

Minimal First-Paper Model

For the first perturbation paper, the clean model is not the full RUT unification architecture. It is a minimal bridge-activated modified-growth/lensing model that can be falsified against data.

Include:

- $B(a)$, the bridge activation profile.
- ξ_μ and ξ_Σ as effective working couplings.
- m_ϕ and m_Σ as effective mass scales, preferably derived or tightly prior-bounded.
- GR correspondence checks: $B \rightarrow 0$ implies $\mu \rightarrow 1$, $\Sigma \rightarrow 1$, $\eta \rightarrow 1$, and $E_G \rightarrow \Omega_{m0}/f(a)$.

Exclude or fix:

- g and $I_{0\text{inf}}$.
- β_{-} , β_{+} , β_x .
- extra ladder/inversion/light-time parameters.
- free Q-sector amplitude unless the quadratic-action stability and anisotropic-stress signs are explicitly checked.

Paper-Safe Wording

The first RUT perturbation paper should say that μ and Σ are effective, bridge-activated test functions motivated by the quasi-static matrix reduction. Their coefficients are not fundamental new constants. They represent combinations of the action-level couplings and background fields that must be derived in the next coefficient-level closure step.

Recommended claim level:

RUT currently supplies a structural derivation of the quasi-static response form; the exact coefficients remain open until the full quadratic perturbation action is solved.

Step 10 Conclusion

Step 10 narrows RUT into a defensible first-paper model. The purpose is not to include every promising idea, but to test the smallest bridge-activated perturbation sector that connects background expansion, growth, lensing, gravitational slip, and E_G . Parameters with high overfitting risk should be fixed, excluded, or moved to future work.

RUT Perturbation Sprint - Step 11

Unresolved Derivation Targets for the RUT Framework

This step summarizes the principal mathematical and phenomenological gaps that remain to be closed in the current Reverse Universe Theory (RUT) formulation. Each entry indicates the area of work, its status, and why it matters. Items labelled **[O]** are open derivations; **[F]** points to tests or falsifiers once the derivation is completed; **[I]** marks interpretive material better suited to the Master book or later theoretical development.

Topic / Area	Description & Tasks	Status / Label
Q-sector sign and stress-energy tensor	Complete the variation of the Q-sector Lagrangian $L_Q = -1/2 \xi_Q B(\chi) (\nabla_\mu \theta \nabla^\mu \phi)^2$ to obtain an unambiguous expression for $T_{\mu\nu}(Q)$. Check the sign of the background energy density and pressure and derive the linear anisotropic-stress component. This determines whether the Q-sector can support $\Phi \neq \Psi$.	[O], [D] in progress
Explicit mass matrices and couplings	Evaluate the effective mass matrix $M(k,a)$ and coupling vectors C_A , D_A , and α_B from the quadratic action for the coupled fields (ϕ, χ, θ) . This requires second derivatives of $K(B)$, $V(\phi, B)$, $U(\chi)$, plus $B_{,\chi}$ and cross terms. The coefficients m_ϕ , m_χ , m_θ , m_Σ and ξ_μ , ξ_Σ follow from this.	[O]
Full quasi-static reduction	Solve the algebraic system $M(k,a) \delta \varphi = S$ under $k \gg aH$ and inside the relevant sound horizon, then substitute back into the Poisson and Weyl equations. Derive explicit formulas for $\mu(k,a)$ and $\Sigma(k,a)$ in terms of action parameters. Check the scale dependence and match to the working form $1 + \xi B k^2/(k^2 + a^2 m^2)$.	[O]
Gauge-sector anisotropic stress	Determine whether the modulated gauge term $-1/4 Z(B) F_{\mu\nu} F^{\mu\nu}$ contributes anisotropic stress at linear order. If so, compute its contribution to $\Phi - \Psi$. Decide whether this sector can be neglected in the first paper or must be included.	[O]
Backreaction and magneto-acoustic sector	Quantify the effect of the magneto-acoustic sector on the drag horizon and sound speed. Compute the statistical correlators $\langle F_+^2 \rangle$, $\langle F_-^2 \rangle$, and cross terms; check whether β_+ , β_- , and β_x can be derived from first principles using MHD theory.	[O]
Stability and ghost-free conditions	Establish positivity and stability conditions for the kinetic matrix. Derive no-ghost and gradient-stability conditions in the scalar sector, for example $K(B) > 0$ and positive sound speeds squared.	[O], [F]
Comparison of quasi-static vs. full perturbations	Derive the full time-dependent linear perturbation equations beyond the quasi-static approximation and check at which scales the QS approximation breaks down. This is necessary for large-scale, low-k, ISW-scale, or horizon-scale claims.	[O]
Numerical forecast pipeline	Implement a numerical solver for the derived $\mu(k,a)$ and $\Sigma(k,a)$. Compute the growth rate $f(k,a)$, lensing potential, and $E_G(k,a)$ for representative parameter sets. This forecast will identify where RUT deviates from Λ CDM and test model viability.	[O], [F] once simulation is run
Data confrontation and parameter fitting	Develop a likelihood analysis using current cosmological data: BAO, SNe, growth, lensing, and E_G . Fit the minimal RUT parameters and test whether a single parameter set improves all probes simultaneously.	[O]
Microphysical origin of the bridge	Beyond the scalar-field implementation, explore whether the tanh bridge arises from a more fundamental symmetry or UV completion, potentially connecting to high-energy physics or string-theory-like dualities.	[I] - outside first paper scope
Derivation of $I_{0\infty}$, g, and ESZI ladder	The geometry factor g and inversion operator $I_{0\infty}$ are currently conceptual. Provide an action-level derivation or prove they follow from boundary conditions, topology, or compactification. This remains Master-book material rather than first-paper material.	[I]/[O]

Comments

- Items labelled [O] are open derivations required for full mathematical closure. Some may be deferred beyond the first journal paper if they do not directly impact the minimal cosmological test sector.
- Items labelled [F] become falsifiers once the mathematics is worked out and can be compared to data; stability conditions and E_G forecasts are examples.
- Items labelled [I] are interpretive or long-form Master-book material. They should not be placed at the center of the first physics-oriented framework paper unless a derivation is supplied.
- First-paper discipline: prioritize the minimal perturbation/growth/lensing sector. Defer high-risk parameters such as g , $I_{0\infty}$, β_+ , β_- , and β_x until they are either derived or explicitly tested.